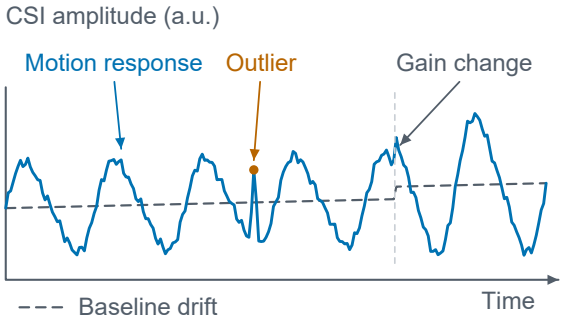
(a) Raw CSI contains mixed effects

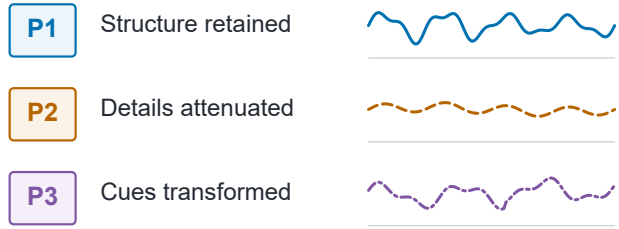


Amplitude: gain variation, drift, and outliers
Phase: offsets and other distortions

(b) Preprocessing changes retained cues

Denoising / Outliers / Phase calibration
Normalization / Resampling

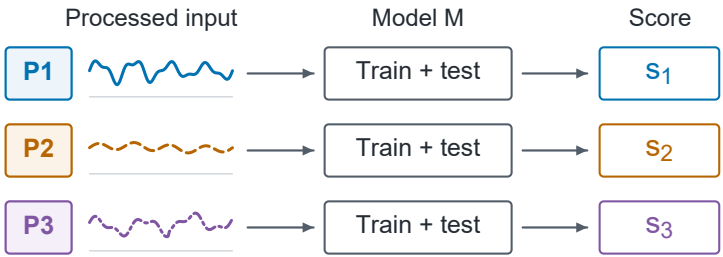
Vary methods, order, and parameters



Possible outcomes; schematic traces

(c) Compare within a fixed architecture

Same data split and training protocol
Same architecture; trained separately



Different inputs may yield different scores.

(d) Model-dependent effects

Feature extraction differs across models.
Compare every preprocessing-model pair:

Input	Model A	Model B
P1	$s_{1,A}$	$s_{1,B}$
P2	$s_{2,A}$	$s_{2,B}$
P3	$s_{3,A}$	$s_{3,B}$

> Preprocessing effects may depend on the model used.

Conceptual illustration: synthetic waveforms and symbolic scores.